

	$\mathcal{K}_{0^+}^{\#1}$	$\mathcal{K}_{0^+}^{\#1}{}_{\alpha\beta\chi}$				
$\mathcal{K}_{0^+}^{\#1}{}^\dagger$	0	0				
$\mathcal{K}_{0^+}^{\#1}{}^\dagger{}^{\alpha\beta\chi}$	0	0	$\mathcal{K}_{1^+}^{\#1}{}_{\alpha\beta}$	$\mathcal{K}_{1^+}^{\#2}{}_{\alpha\beta}$	$\mathcal{K}_{1^+}^{\#1}{}_{\alpha}$	$\mathcal{K}_{1^+}^{\#2}{}_{\alpha}$
$\mathcal{K}_{1^+}^{\#1}{}^\dagger{}^{\alpha\beta}$	$-\frac{\Upsilon_1 k^2}{2}$	$\frac{k^2}{2} \frac{{}^{(4)}\kappa_4}{\sqrt{2}}$	0	0		
$\mathcal{K}_{1^+}^{\#2}{}^\dagger{}^{\alpha\beta}$	$\frac{k^2}{2} \frac{{}^{(4)}\kappa_4}{\sqrt{2}}$	$\frac{\Upsilon_2 k^2}{2}$	0	0		
$\mathcal{K}_{1^+}^{\#1}{}^\dagger{}^{\alpha}$	0	0	0	0		
$\mathcal{K}_{1^+}^{\#2}{}^\dagger{}^{\alpha}$	0	0	0	0	$\mathcal{K}_{2^+}^{\#1}{}_{\alpha\beta}$	$\mathcal{K}_{2^+}^{\#1}{}_{\alpha\beta\chi}$
			$\mathcal{K}_{2^+}^{\#1}{}^\dagger{}^{\alpha\beta}$	$\frac{3 k^2}{2} \frac{{}^{(4)}\kappa_1}{\sqrt{2}}$	0	
			$\mathcal{K}_{2^+}^{\#1}{}^\dagger{}^{\alpha\beta\chi}$	0	0	

	$\mathcal{T}_{0^+}^{\#1}$	$\mathcal{T}_{0^+}^{\#1}{}_{\alpha\beta\chi}$				
$\mathcal{T}_{0^+}^{\#1}{}^\dagger$	0	0				
$\mathcal{T}_{0^+}^{\#1}{}^{\dagger\alpha\beta\chi}$	0	0	$\mathcal{T}_{1^+}^{\#1}{}_{\alpha\beta}$	$\mathcal{T}_{1^+}^{\#2}{}_{\alpha\beta}$	$\mathcal{T}_{1^+}^{\#1}{}_{\alpha}$	$\mathcal{T}_{1^+}^{\#2}{}_{\alpha}$
$\mathcal{T}_{1^+}^{\#1}{}^\dagger{}^{\alpha\beta}$	$\frac{\Upsilon_2 k^2}{2 \text{Det}(1^+)}$	$-\frac{k^2 \kappa_4^{(4)}}{2 \sqrt{2} \text{Det}(1^+)}$	0	0		
$\mathcal{T}_{1^+}^{\#2}{}^\dagger{}^{\alpha\beta}$	$-\frac{k^2 \kappa_4^{(4)}}{2 \sqrt{2} \text{Det}(1^+)}$	$-\frac{\Upsilon_1 k^2}{2 \text{Det}(1^+)}$	0	0		
$\mathcal{T}_{1^+}^{\#1}{}^\dagger{}^{\alpha}$	0	0	0	0		
$\mathcal{T}_{1^+}^{\#2}{}^\dagger{}^{\alpha}$	0	0	0	0	$\mathcal{T}_{2^+}^{\#1}{}_{\alpha\beta}$	$\mathcal{T}_{2^+}^{\#1}{}_{\alpha\beta\chi}$
			$\mathcal{T}_{2^+}^{\#1}{}^\dagger{}^{\alpha\beta}$	$\frac{2}{3 k^2 \kappa_1^{(4)}}$	0	
			$\mathcal{T}_{2^+}^{\#1}{}^{\dagger\alpha\beta\chi}$	0	0	

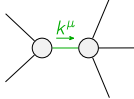
Abbreviations used in matrices

$$\Upsilon_1 == 3 \frac{{}^{(4)}\kappa_1}{\kappa_1} + 2 \frac{{}^{(4)}\kappa_3}{\kappa_3} \&\& \Upsilon_2 == \frac{{}^{(4)}\kappa_3}{\kappa_3} + \frac{{}^{(4)}\kappa_4}{\kappa_4} \&\& \text{Det}(1^+) == -\frac{1}{8} k^4 (6 \frac{{}^{(4)}\kappa_1}{\kappa_1} (\frac{{}^{(4)}\kappa_3}{\kappa_3} + \frac{{}^{(4)}\kappa_4}{\kappa_4}) + (2 \frac{{}^{(4)}\kappa_3}{\kappa_3} + \frac{{}^{(4)}\kappa_4}{\kappa_4})^2)$$

Lagrangian

$$\frac{{}^{(4)}\kappa_1}{\kappa_1} \partial_\beta \mathcal{K}^\delta_{\chi\delta} \partial^\chi \mathcal{K}^\alpha_{\alpha}{}^\beta + \frac{{}^{(4)}\kappa_3}{\kappa_3} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\alpha\chi}{}^\delta + \frac{{}^{(4)}\kappa_4}{\kappa_4} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}{}^\delta -$$

$$3 \frac{{}^{(4)}\kappa_1}{\kappa_1} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\chi\alpha}{}^\delta - \frac{{}^{(4)}\kappa_3}{\kappa_3} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\chi\alpha}{}^\delta + \frac{1}{2} \frac{{}^{(4)}\kappa_3}{\kappa_3} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}{}^\delta + \frac{1}{2} \frac{{}^{(4)}\kappa_4}{\kappa_4} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}{}^\delta$$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi} \mathcal{J}_{\alpha\beta\chi}$		
Source constraint(s)	# constraint(s)	Covariant form	
$\mathcal{J}_0^{\#1\alpha\beta\chi} == 0$	1	$\partial_\delta \partial^\alpha \mathcal{J}^{\beta\chi\delta} + \partial_\delta \partial^\alpha \mathcal{J}^{\delta\beta\chi} + \partial_\delta \partial^\beta \mathcal{J}^{\chi\alpha\delta} + \partial_\delta \partial^\chi \mathcal{J}^{\alpha\beta\delta} + \partial_\delta \partial^\chi \mathcal{J}^{\delta\alpha\beta} + \partial_\delta \partial^\delta \mathcal{J}^{\beta\alpha\chi} ==$ $\partial_\delta \partial^\alpha \mathcal{J}^{\chi\beta\delta} + \partial_\delta \partial^\beta \mathcal{J}^{\alpha\chi\delta} + \partial_\delta \partial^\beta \mathcal{J}^{\delta\alpha\chi} + \partial_\delta \partial^\chi \mathcal{J}^{\beta\alpha\delta} + \partial_\delta \partial^\delta \mathcal{J}^{\alpha\beta\chi} + \partial_\delta \partial^\delta \mathcal{J}^{\chi\alpha\beta}$	
$\mathcal{J}_{0^+}^{\#1} == 0$	1	$\partial_\beta \mathcal{J}^\alpha{}^\beta{}_\alpha == 0$	
$\mathcal{J}_{1^+}^{\#2\alpha} == 0$	3	$\partial_\chi \partial_\beta \mathcal{J}^{\beta\alpha\chi} == 0$	
$\mathcal{J}_{1^+}^{\#1\alpha} == 0$	3	$\partial_\chi \partial^\alpha \mathcal{J}^\beta{}_\beta{}^\chi + \partial_\chi \partial^\chi \mathcal{J}^{\beta\alpha}{}_\beta == \partial_\chi \partial_\beta \mathcal{J}^{\beta\alpha\chi}$	
$\mathcal{J}_2^{\#1\alpha\beta\chi} == 0$	5	$3 \partial_\epsilon \partial_\delta \partial^\chi \partial^\alpha \mathcal{J}^{\delta\beta\epsilon} + 3 \partial_\epsilon \partial^\epsilon \partial^\chi \partial^\alpha \mathcal{J}^{\delta\beta}{}_\delta + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\beta \mathcal{J}^{\alpha\chi\delta} + 4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\beta \mathcal{J}^{\chi\alpha\delta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\beta \mathcal{J}^{\delta\alpha\chi} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\chi \mathcal{J}^{\beta\alpha\delta} +$ $4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\chi \mathcal{J}^{\delta\alpha\beta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\delta \mathcal{J}^{\alpha\beta\chi} + 3 \eta^{\beta\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\alpha \mathcal{J}^\delta{}_\delta{}^\epsilon + 3 \eta^{\alpha\chi} \partial_\phi \partial^\phi \partial_\epsilon \mathcal{J}^\delta \mathcal{J}^{\delta\beta\epsilon} + 3 \eta^{\beta\chi} \partial_\phi \partial^\phi \partial_\epsilon \mathcal{J}^\delta \mathcal{J}^{\delta\alpha}{}_\delta ==$ $3 \partial_\epsilon \partial_\delta \partial^\chi \partial^\beta \mathcal{J}^{\delta\alpha\epsilon} + 3 \partial_\epsilon \partial^\epsilon \partial^\chi \partial^\beta \mathcal{J}^{\delta\alpha}{}_\delta + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\beta\chi\delta} + 4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\chi\beta\delta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\delta\beta\chi} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\alpha\beta\delta} +$ $2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\delta \mathcal{J}^{\beta\alpha\chi} + 4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\delta \mathcal{J}^{\chi\alpha\beta} + 3 \eta^{\alpha\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\beta \mathcal{J}^\delta{}_\delta{}^\epsilon + 3 \eta^{\beta\chi} \partial_\phi \partial^\phi \partial_\epsilon \mathcal{J}^\delta \mathcal{J}^{\delta\alpha\epsilon} + 3 \eta^{\alpha\chi} \partial_\phi \partial^\phi \partial_\epsilon \mathcal{J}^{\delta\beta}{}_\delta$	
Total # constraint(s):	13		
Resolved pole(s)	# polarization(s)	Square mass	Residue
	2	0	$\frac{24 \frac{{}^{(4)}\kappa_1}{\kappa_1}{}^2 + 8 \frac{{}^{(4)}\kappa_1}{\kappa_1} (2 \frac{{}^{(4)}\kappa_3}{\kappa_3} + \frac{{}^{(4)}\kappa_4}{\kappa_4}) + (2 \frac{{}^{(4)}\kappa_3}{\kappa_3} + \frac{{}^{(4)}\kappa_4}{\kappa_4})^2}{\frac{{}^{(4)}\kappa_1}{\kappa_1} (6 \frac{{}^{(4)}\kappa_1}{\kappa_1} (\frac{{}^{(4)}\kappa_3}{\kappa_3} + \frac{{}^{(4)}\kappa_4}{\kappa_4}) + (2 \frac{{}^{(4)}\kappa_3}{\kappa_3} + \frac{{}^{(4)}\kappa_4}{\kappa_4})^2)}$

Resolved unitarity condition(s):

$$(\frac{{}^{(4)}\kappa_4}{\kappa_4} < 0 \&\& ((\frac{{}^{(4)}\kappa_3}{\kappa_3} < -\frac{{}^{(4)}\kappa_4}{2} \&\& 0 < \frac{{}^{(4)}\kappa_1}{\kappa_1} < \frac{-4 \frac{{}^{(4)}\kappa_3}{\kappa_3}{}^2 - 4 \frac{{}^{(4)}\kappa_3}{\kappa_3} \frac{{}^{(4)}\kappa_4}{\kappa_4} - \frac{{}^{(4)}\kappa_4}{}^2}{6 \frac{{}^{(4)}\kappa_3}{\kappa_3} + 6 \frac{{}^{(4)}\kappa_4}{\kappa_4}})) \parallel (-\frac{{}^{(4)}\kappa_4}{2} < \frac{{}^{(4)}\kappa_3}{\kappa_3} < -\frac{{}^{(4)}\kappa_4}{\kappa_4} \&\& 0 < \frac{{}^{(4)}\kappa_1}{\kappa_1} < \frac{-4 \frac{{}^{(4)}\kappa_3}{\kappa_3}{}^2 - 4 \frac{{}^{(4)}\kappa_3}{\kappa_3} \frac{{}^{(4)}\kappa_4}{\kappa_4} - \frac{{}^{(4)}\kappa_4}{}^2}{6 \frac{{}^{(4)}\kappa_3}{\kappa_3} + 6 \frac{{}^{(4)}\kappa_4}{\kappa_4}})) \parallel$$

$$(\frac{{}^{(4)}\kappa_3}{\kappa_3} == -\frac{{}^{(4)}\kappa_4}{\kappa_4} \&\& \frac{{}^{(4)}\kappa_1}{\kappa_1} > 0) \parallel ((\frac{{}^{(4)}\kappa_3}{\kappa_3} > -\frac{{}^{(4)}\kappa_4}{\kappa_4} \&\& (\frac{{}^{(4)}\kappa_1}{\kappa_1} < \frac{-4 \frac{{}^{(4)}\kappa_3}{\kappa_3}{}^2 - 4 \frac{{}^{(4)}\kappa_3}{\kappa_3} \frac{{}^{(4)}\kappa_4}{\kappa_4} - \frac{{}^{(4)}\kappa_4}{}^2}{6 \frac{{}^{(4)}\kappa_3}{\kappa_3} + 6 \frac{{}^{(4)}\kappa_4}{\kappa_4}} \parallel \frac{{}^{(4)}\kappa_1}{\kappa_1} > 0)))) \parallel$$

$$(\frac{{}^{(4)}\kappa_4}{\kappa_4} == 0 \&\& ((\frac{{}^{(4)}\kappa_3}{\kappa_3} < 0 \&\& 0 < \frac{{}^{(4)}\kappa_1}{\kappa_1} < -\frac{2 \frac{{}^{(4)}\kappa_3}{\kappa_3}}{3}) \parallel (\frac{{}^{(4)}\kappa_3}{\kappa_3} > 0 \&\& (\frac{{}^{(4)}\kappa_1}{\kappa_1} < -\frac{2 \frac{{}^{(4)}\kappa_3}{\kappa_3}}{3} \parallel \frac{{}^{(4)}\kappa_1}{\kappa_1} > 0)))) \parallel$$

$$(\frac{{}^{(4)}\kappa_4}{\kappa_4} > 0 \&\& ((\frac{{}^{(4)}\kappa_3}{\kappa_3} < -\frac{{}^{(4)}\kappa_4}{\kappa_4} \&\& 0 < \frac{{}^{(4)}\kappa_1}{\kappa_1} < \frac{-4 \frac{{}^{(4)}\kappa_3}{\kappa_3}{}^2 - 4 \frac{{}^{(4)}\kappa_3}{\kappa_3} \frac{{}^{(4)}\kappa_4}{\kappa_4} - \frac{{}^{(4)}\kappa_4}{}^2}{6 \frac{{}^{(4)}\kappa_3}{\kappa_3} + 6 \frac{{}^{(4)}\kappa_4}{\kappa_4}})) \parallel$$

$$(\frac{{}^{(4)}\kappa_3}{\kappa_3} == -\frac{{}^{(4)}\kappa_4}{\kappa_4} \&\& \frac{{}^{(4)}\kappa_1}{\kappa_1} > 0) \parallel (-\frac{{}^{(4)}\kappa_4}{\kappa_4} < \frac{{}^{(4)}\kappa_3}{\kappa_3} < -\frac{{}^{(4)}\kappa_4}{2} \&\& (\frac{{}^{(4)}\kappa_1}{\kappa_1} < \frac{-4 \frac{{}^{(4)}\kappa_3}{\kappa_3}{}^2 - 4 \frac{{}^{(4)}\kappa_3}{\kappa_3} \frac{{}^{(4)}\kappa_4}{\kappa_4} - \frac{{}^{(4)}\kappa_4}{}^2}{6 \frac{{}^{(4)}\kappa_3}{\kappa_3} + 6 \frac{{}^{(4)}\kappa_4}{\kappa_4}} \parallel \frac{{}^{(4)}\kappa_1}{\kappa_1} > 0)) \parallel$$

$$(\frac{{}^{(4)}\kappa_3}{\kappa_3} == -\frac{{}^{(4)}\kappa_4}{\kappa_4} \&\& (\frac{{}^{(4)}\kappa_1}{\kappa_1} < 0 \parallel \frac{{}^{(4)}\kappa_1}{\kappa_1} > 0)) \parallel (\frac{{}^{(4)}\kappa_3}{\kappa_3} > -\frac{{}^{(4)}\kappa_4}{2} \&\& (\frac{{}^{(4)}\kappa_1}{\kappa_1} < \frac{-4 \frac{{}^{(4)}\kappa_3}{\kappa_3}{}^2 - 4 \frac{{}^{(4)}\kappa_3}{\kappa_3} \frac{{}^{(4)}\kappa_4}{\kappa_4} - \frac{{}^{(4)}\kappa_4}{}^2}{6 \frac{{}^{(4)}\kappa_3}{\kappa_3} + 6 \frac{{}^{(4)}\kappa_4}{\kappa_4}} \parallel \frac{{}^{(4)}\kappa_1}{\kappa_1} > 0))))$$